

TED (10)-4079/5059
(REVISION – 2010)

Reg. No.
Signature

MODEL QUESTION PAPER
SIXTH SEMESTER DIPLOMA EXAMINATION IN COMPUTER HARDWARE
MAINTENANCE – MARCH, 2013

INFORMATION SECURITY

[Time : 3 hours]

(Maximum marks : 100)

PART – A

Marks

I Answer the following questions in one or two sentences. Each question carries 2 marks.

1. Define Digital signature
2. Write the two common techniques used to protect a password file
3. List the fields in audit record
4. Define Honey Pots
5. State the need of firewall

(5 x 2 = 10)

PART – B

II Answer *any five* of the following. Each question carries 6 marks.

1. Explain symmetric encryption with block diagram
2. Explain the use of hash function
3. With neat diagram explain the operation of biometric authentication system
4. Explain various access control requirements
5. Summarize the logical components of Intrusion Detection System
6. Explain SYN spoofing
7. Describe the defense against DDos

(5 x 6 = 30)

[PTO]

PART – C

(Answer one full question from each unit. Each questions carries 15 marks.)

Unit – I

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| III | (a) | Briefly explain the OSI security architecture along with the security mechanism available | 8 |
| | (b) | Discuss the categories of passive and active network security attacks | 7 |

OR

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| IV | (a) | Describe message authentication with one way hash function | 8 |
| | (b) | Discuss public key encryption structure | 7 |

Unit – II

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| V | (a) | Explain the operation of biometric authentication system | 8 |
| | (b) | Explain the attack against password based authentication | 7 |

OR

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| VI | (a) | Describe various security issues for user identification? | 8 |
| | (b) | Discuss Unix file access control system | 7 |

Unit – III

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| VII | (a) | Explain classification of ROOTKITS | 7 |
| | (b) | Explain architecture of SNORT IDS with diagram | 8 |

OR

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| VIII | (a) | Explain various intrusion Detection techniques | 9 |
| | (b) | Write down the network based intrusion technique | 6 |

Unit – IV

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| IX | (a) | Briefly explain Bastion host | 8 |
| | (b) | Explain network based IPS | 7 |

OR

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| X | (a) | Explain packet filtering firewall | 8 |
| | (b) | State the limitations of firewall | 7 |

